STRIPE AI INFRASTRUCTURE • CERTIFICATION BLOC 2 (RNCP 41993)

Requêtes d'Exploitation & Validation Métier (SQL OLAP & NoSQL DocumentDB)

Les 6 requêtes SQL suivantes exploitent le schéma en étoile (`stripe_olap`) sur **Amazon Redshift Serverless** pour répondre aux questions stratégiques du CFO, des Risk Managers, du Product Management et de la Conformité.

Requête SQL 1 : Chiffre d'Affaires Brut, Net et Commissions Stripe par Marchand et par Mois

- **Question Métier :** Quels sont les 10 marchands générant le plus de volume d'affaires au cours du dernier trimestre, quel est le montant des commissions nettes perçues par Stripe, et quelle est la part des paiements transfrontaliers (*cross-border*) ?
- **Rôle Analytique :** Identification des *Key Accounts* et pilotage de la rentabilité nette par marchand.

```
SELECT
    m.merchant_id,
    m.business_name,
    m.mcc_category_name,
    d.year,
    d.month_name,
    ROUND(SUM(f.gross_amount_cents) / 100.0, 2) AS total_volume_eur,
    ROUND(SUM(f.stripe_fee_cents - COALESCE(f.interchange_fee_cents, 0)) / 100.0, 2) AS net_stripe
    COUNT(f.fact_txn_key) AS total_transactions,
    ROUND(100.0 * SUM(CASE WHEN pm.is_cross_border THEN f.gross_amount_cents ELSE 0 END) /
          NULLIF(SUM(f.gross_amount_cents), 0), 2) AS cross_border_volume_pct
FROM stripe_olap.fact_transactions f
JOIN stripe_olap.dim_merchant m ON f.merchant_key = m.merchant_key
JOIN stripe_olap.dim_date d ON f.date_key = d.date_key
JOIN stripe_olap.dim_payment_method pm ON f.payment_method_key = pm.payment_method_key
JOIN stripe_olap.dim_status s ON f.status_key = s.status_key
WHERE s.payment_status = 'succeeded'
  AND d.year = 2026
  AND d.quarter = 3
GROUP BY m.merchant_id, m.business_name, m.mcc_category_name, d.year, d.month_name
ORDER BY total_volume_eur DESC
LIMIT 10;
```

Requête SQL 2 : Taux d'Échec et Motifs de Refus par Moyen de Paiement et Pays Émetteur

- **Question Métier :** Quels corridors de paiement présentent un taux de refus anormalement élevé (> 8%), et quelles sont les causes dominantes (fonds insuffisants vs blocage fraude) ?
- **Rôle Analytique :** Optimisation du routage intelligent et calibration de **Smart Retries**.

```
SELECT
    pm.type AS payment_method_type,
    pm.card_brand,
    geo.country_code AS issuing_country,
    COUNT(f.fact_txn_key) AS total_attempts,
    ROUND(100.0 * SUM(CASE WHEN s.payment_status = 'failed' THEN 1 ELSE 0 END) /
          NULLIF(COUNT(f.fact_txn_key), 0), 2) AS failure_rate_pct,
    SUM(CASE WHEN s.failure_code = 'insufficient_funds' THEN 1 ELSE 0 END) AS count_insufficient_f
    SUM(CASE WHEN s.failure_code = 'fraud_blocked' THEN 1 ELSE 0 END) AS count_fraud_blocked
FROM stripe_olap.fact_transactions f
JOIN stripe_olap.dim_payment_method pm ON f.payment_method_key = pm.payment_method_key
JOIN stripe_olap.dim_geo_location geo ON f.geo_key = geo.geo_key
JOIN stripe_olap.dim_status s ON f.status_key = s.status_key
JOIN stripe_olap.dim_date d ON f.date_key = d.date_key
WHERE d.full_date ⩾ CURRENT_DATE - INTERVAL '30 days'
GROUP BY pm.type, pm.card_brand, geo.country_code
HAVING COUNT(f.fact_txn_key) > 1000
   AND 100.0 * SUM(CASE WHEN s.payment_status = 'failed' THEN 1 ELSE 0 END) / COUNT(f.fact_txn_key
ORDER BY failure_rate_pct DESC;
```

Requête SQL 3 : Analyse des Contestations (Chargebacks) et Seuil d'Alerte Visa/Mastercard (1%)

- **Question Métier :** Quels marchands approchent ou dépassent le seuil critique de 0.75% / 1% de contestations sur les 30 derniers jours ?
- **Rôle Analytique :** Surveillance du risque crédit et prévention des sanctions *Card Schemes*.

```
SELECT
    m.merchant_id,
    m.business_name,
    m.mcc_category_name,
    COUNT(f.fact_txn_key) AS total_transactions,
    SUM(f.is_disputed) AS total_chargebacks,
    ROUND(100.0 * SUM(f.is_disputed) / NULLIF(COUNT(f.fact_txn_key), 0), 3) AS chargeback_rate_pct
    ROUND(SUM(CASE WHEN f.is_disputed = 1 THEN f.gross_amount_cents ELSE 0 END) / 100.0, 2) AS dis
    CASE
        WHEN (100.0 * SUM(f.is_disputed) / COUNT(f.fact_txn_key)) ⩾ 1.0 THEN 'CRITICAL - IMMEDIAT
        WHEN (100.0 * SUM(f.is_disputed) / COUNT(f.fact_txn_key)) ⩾ 0.75 THEN 'WARNING - ENHANCED
        ELSE 'NORMAL'
    END AS risk_enforcement_status
FROM stripe_olap.fact_transactions f
JOIN stripe_olap.dim_merchant m ON f.merchant_key = m.merchant_key
JOIN stripe_olap.dim_date d ON f.date_key = d.date_key
WHERE d.full_date ⩾ CURRENT_DATE - INTERVAL '30 days'
GROUP BY m.merchant_id, m.business_name, m.mcc_category_name
HAVING COUNT(f.fact_txn_key) ⩾ 500
   AND (100.0 * SUM(f.is_disputed) / COUNT(f.fact_txn_key)) ⩾ 0.75
ORDER BY chargeback_rate_pct DESC;
```

Requête SQL 4 : Analyse de Cohorte et Taux d'Attrition (*Churn*) Mensuel des Marchands

- **Question Métier :** Quel est le taux de rétention des nouvelles cohortes de marchands inscrits au cours des 6 derniers mois ?
- **Rôle Analytique :** Évaluation de la valeur vie client (LTV) et du taux de rétention B2B.

```
WITH MerchantCohort AS (
    SELECT
        merchant_key,
        MIN(date_key / 100) AS cohort_month
    FROM stripe_olap.fact_daily_merchant_revenue
    GROUP BY merchant_key
),
MonthlyActivity AS (
    SELECT
        f.merchant_key,
        c.cohort_month,
        (f.date_key / 100) AS activity_month,
        SUM(f.total_volume_cents) / 100.0 AS merchant_monthly_volume
    FROM stripe_olap.fact_daily_merchant_revenue f
    JOIN MerchantCohort c ON f.merchant_key = c.merchant_key
    GROUP BY f.merchant_key, c.cohort_month, (f.date_key / 100)
)
SELECT
    cohort_month,
    activity_month,
    COUNT(DISTINCT merchant_key) AS active_merchants_count,
    ROUND(AVG(merchant_monthly_volume), 2) AS avg_volume_per_merchant_eur
FROM MonthlyActivity
GROUP BY cohort_month, activity_month
ORDER BY cohort_month, activity_month;
```

Requête SQL 5 : Segmentation Client RFM (Récence, Fréquence, Montant) & Respect CCPA Opt-Out

- **Question Métier :** Comment se répartissent nos clients finaux par valeur comportementale (RFM) et zone réglementaire, tout en excluant strictement les profils ayant activé le refus de partage CCPA pour les campagnes marketing ?
- **Rôle Analytique :** Segmentation client exploitable par les équipes Produit et Growth sans infraction réglementaire.

```
WITH CustomerMetrics AS (
    SELECT
        c.customer_key,
        c.risk_segment,
        c.rfm_tier,
        geo.regulatory_zone,
        MAX(d.full_date) AS last_purchase_date,
        COUNT(f.fact_txn_key) AS frequency_txns,
        ROUND(SUM(f.gross_amount_cents) / 100.0, 2) AS monetary_eur
    FROM stripe_olap.fact_transactions f
    JOIN stripe_olap.dim_customer c ON f.customer_key = c.customer_key
    JOIN stripe_olap.dim_geo_location geo ON f.geo_key = geo.geo_key
    JOIN stripe_olap.dim_date d ON f.date_key = d.date_key
    JOIN stripe_olap.dim_status s ON f.status_key = s.status_key
    WHERE s.payment_status = 'succeeded'
      AND c.do_not_sell_or_share = FALSE  -- Filtrage strict CCPA Opt-Out
      AND d.full_date ⩾ CURRENT_DATE - INTERVAL '180 days'
    GROUP BY c.customer_key, c.risk_segment, c.rfm_tier, geo.regulatory_zone
)
SELECT
    regulatory_zone,
    rfm_tier,
    risk_segment,
    COUNT(*) AS customer_count,
    ROUND(AVG(frequency_txns), 1) AS avg_frequency,
    ROUND(AVG(monetary_eur), 2) AS avg_spend_eur,
    ROUND(SUM(monetary_eur), 2) AS segment_total_volume_eur
FROM CustomerMetrics
GROUP BY regulatory_zone, rfm_tier, risk_segment
ORDER BY segment_total_volume_eur DESC;
```

Requête SQL 6 : Suivi de Performance Produit & Revenus Récurrents (Abonnements)

- **Question Métier :** Quelle part du volume total provient des offres par abonnement (`recurring`) comparée aux ventes unitaires (`one_time`) par catégorie produit ?
- **Rôle Analytique :** Pilotage stratégique des produits SaaS/Stripe Billing et mesure de la résilience des revenus récurrents.

```
SELECT
    p.category AS product_category,
    p.billing_type,
    COUNT(DISTINCT f.merchant_key) AS active_merchants,
    COUNT(f.fact_txn_key) AS txns_volume_count,
    ROUND(SUM(f.gross_amount_cents) / 100.0, 2) AS total_gpv_eur,
    ROUND(100.0 * SUM(CASE WHEN f.is_recurring = 1 THEN f.gross_amount_cents ELSE 0 END) /
          NULLIF(SUM(f.gross_amount_cents), 0), 2) AS recurring_revenue_share_pct
FROM stripe_olap.fact_transactions f
JOIN stripe_olap.dim_product p ON f.product_key = p.product_key
JOIN stripe_olap.dim_status s ON f.status_key = s.status_key
JOIN stripe_olap.dim_date d ON f.date_key = d.date_key
WHERE s.payment_status = 'succeeded'
  AND d.year = 2026
GROUP BY p.category, p.billing_type
ORDER BY total_gpv_eur DESC;
```

---

2. Requêtes NoSQL Documentaires (MongoDB / Amazon DocumentDB)

Les 5 pipelines d'agrégation suivants exploitent nos collections documentaires pour la sécurité temps réel, le monitoring technique, l'analyse NLP et la couche ML.

Requête NoSQL 1 : Détection d'Attaque par Vélocité Matérielle (*Device Fingerprint Velocity*)

- **Question Métier :** Quels terminaux ont tenté d'utiliser plus de 3 cartes bancaires distinctes au cours des 24 dernières heures (*Carding Attack*) ?

```
db.device_fingerprints.aggregate([
  {
    $match: {
      "last_seen": { $gte: new Date(Date.now() - 24 * 60 * 60 * 1000) },
      "velocity_metrics.cards_associated_count": { $gte: 3 }
    }
  },
  {
    $project: {
      _id: 1,
      fingerprint_hash: 1,
      customer_id: 1,
      cards_tested_count: "$velocity_metrics.cards_associated_count",
      merchants_hit: "$velocity_metrics.merchants_visited_24h",
      is_bot: "$bot_indicators.is_headless",
      latest_ip: { $arrayElemAt: ["$ip_addresses.ip", -1] }
    }
  },
  { $sort: { cards_tested_count: -1, merchants_hit: -1 } },
  { $limit: 50 }
]);
```

Requête NoSQL 2 : Détection de Comportement Robotique (*Bot Checkout Detection*)

- **Question Métier :** Quels marchands subissent actuellement des salves d'attaques robotisées (saisie CVV < 200 ms ou entropie souris artificielle) sur leurs formulaires Checkout ?

```
db.clickstream_events.aggregate([
  {
    $match: {
      "timestamp": { $gte: new Date(Date.now() - 6 * 60 * 60 * 1000) },
      $or: [
        { "form_interaction.cvv_typing_duration_ms": { $lt: 200 } },
        { "form_interaction.mouse_entropy_score": { $lt: 0.15 } },
        { "risk_evaluation.behavioral_bot_score": { $gte: 0.85 } }
      ]
    }
  },
  {
    $group: {
      _id: "$merchant_id",
      bot_attempts_count: { $sum: 1 },
      avg_bot_score: { $avg: "$risk_evaluation.behavioral_bot_score" },
      sample_ips: { $addToSet: "$customer_ip" }
    }
  },
  { $match: { bot_attempts_count: { $gte: 10 } } },
  { $sort: { bot_attempts_count: -1 } }
]);
```

Requête NoSQL 3 : Analyse des Échecs de Webhooks & Latence Réseau

- **Question Métier :** Quels marchands rencontrent un pic d'échecs de distribution de webhooks (erreurs HTTP 5xx ou timeouts > 5s) au cours de la dernière heure ?

```
db.webhook_delivery_logs.aggregate([
  {
    $match: {
      "created_at": { $gte: new Date(Date.now() - 60 * 60 * 1000) },
      "delivery_status": { $ne: "succeeded" }
    }
  },
  { $unwind: "$attempts" },
  {
    $group: {
      _id: { merchant_id: "$merchant_id", endpoint: "$endpoint_url" },
      failed_events_count: { $sum: 1 },
      avg_latency_ms: { $avg: "$attempts.latency_ms" },
      http_error_codes: { $addToSet: "$attempts.http_status" }
    }
  },
  { $match: { failed_events_count: { $gte: 5 } } },
  { $sort: { failed_events_count: -1 } }
]);
```

Requête NoSQL 4 : Analyse NLP des Retours Clients & Détection de Signaux de Churn (`customer_feedback`)

- **Question Métier :** Quels sujets d'irritation (`detected_topics`) génèrent le plus de retours négatifs (NPS <= 6) et de signaux d'attrition au cours des 30 derniers jours ?
- **Résultat Attendu :** Priorisation de la roadmap Produit (ex: réduction de la friction 3D-Secure mobile) et déclenchement proactif de l'équipe Support.

```
db.customer_feedback.aggregate([
  {
    $match: {
      "created_at": { $gte: new Date(Date.now() - 30 * 24 * 60 * 60 * 1000) },
      "ccpa_opt_out": false,
      "nlp_enrichment.sentiment_score": { $lt: -0.25 }
    }
  },
  { $unwind: "$nlp_enrichment.detected_topics" },
  {
    $group: {
      _id: "$nlp_enrichment.detected_topics",
      complaints_count: { $sum: 1 },
      avg_nps: { $avg: "$nps_score" },
      avg_sentiment: { $avg: "$nlp_enrichment.sentiment_score" },
      impacted_merchants: { $addToSet: "$merchant_id" }
    }
  },
  {
    $project: {
      topic: "$_id",
      complaints_count: 1,
      avg_nps: { $round: ["$avg_nps", 2] },
      avg_sentiment: { $round: ["$avg_sentiment", 3] },
      impacted_merchants_count: { $size: "$impacted_merchants" }
    }
  },
  { $sort: { complaints_count: -1 } }
]);
```

Requête NoSQL 5 : Extraction de Caractéristiques ML & Explicabilité SHAP (`ml_feature_vectors`)

- **Question Métier :** Comment auditer les facteurs explicatifs SHAP dominants ayant justifié le blocage ou l'alerte sur les transactions d'un marchand au cours des dernières 24 heures ?

```
db.ml_feature_vectors.aggregate([
  {
    $match: {
      "merchant_id": "acct_1H2X92jks92",
      "computed_at": { $gte: new Date(Date.now() - 24 * 60 * 60 * 1000) },
      "dense_features.device_bot_score": { $gte: 0.50 }
    }
  },
  { $unwind: "$explainability_top_drivers" },
  {
    $group: {
      _id: "$explainability_top_drivers.feature",
      occurrences: { $sum: 1 },
      mean_shap_impact: { $avg: "$explainability_top_drivers.shap_contribution" }
    }
  },
  { $sort: { mean_shap_impact: -1 } }
]);
```